

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

MACMENTA 10 (MEMANTINE HYDROCHLORIDE 10 MG FILM-COATED TABLETS)

MACMENTA 20 (MEMANTINE HYDROCHLORIDE 20 MG FILM-COATED TABLETS)

COMPOSITION:

Each tablet contains:

MACMENTA 10: Memantine hydrochloride10mg (equivalent to 8.31mg Memantine)

MACMENTA 20: Memantine hydrochloride20mg (equivalent to 16.62mg Memantine)

DESCRIPTION:

MACMENTA 10: White to off white, capsule shape, film coated tablet debossed with “CL29” on one side and breakline on both sides ($12.1 \pm 0.3 \times 5.1 \pm 0.3$)

MACMENTA 20: Brown colour, capsule shape, film coated tablet debossed with “CL30” on one side and plain on both sides ($14.6 \pm 0.3 \times 7.1 \pm 0.3$)

PHARMACOLOGICAL ACTION:

Pharmacokinetics:

Absorption

Memantine has an absolute bioavailability of approximately 100%. T_{max} is between 3 and 8 hours. There is no indication that food influences the absorption of memantine.

Distribution

Daily doses of 20 mg lead to steady-state plasma concentrations of memantine ranging from 70 to 150 ng/ml (0.5 - 1 µmol) with large interindividual variations. When daily doses of 5 to 30 mg were administered, a mean cerebrospinal fluid (CSF)/serum ratio of 0.52 was calculated. The volume of distribution is around 10 l/kg. About 45% of memantine is bound to plasma-proteins.

MACMENTA 10 (MEMANTINE HYDROCHLORIDE 10 MG FILM-COATED TABLETS)

MACMENTA 20 (MEMANTINE HYDROCHLORIDE 20 MG FILM-COATED TABLETS)

Biotransformation

In man, about 80% of the circulating memantine-related material is present as the parent compound. Main human metabolites are N-3,5-dimethyl-gludantan, the isomeric mixture of 4- and 6-hydroxy-memantine, and 1-nitroso-3,5-dimethyl-adamantane. None of these metabolites exhibit NMDA-antagonistic activity. No cytochrome P 450 catalysed metabolism has been detected in vitro.

Elimination

Memantine is eliminated in a monoexponential manner with a terminal $t_{1/2}$ of 60 to 100 hours. In patients with normal kidney function, total clearance (Cl_{tot}) amounts to 170 ml/min/1.73 m² and part of total renal clearance is achieved by tubular secretion.

Renal handling also involves tubular reabsorption, probably mediated by cation transport proteins. The renal elimination rate of memantine under alkaline urine conditions may be reduced by a factor of 7 to 9 (see section warning and precaution). Alkalisiation of urine may result from drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or from the massive ingestion of alkalisating gastric buffers.

Linearity

It was demonstrated linear pharmacokinetics in the dose range of 10 to 40 mg.

Pharmacokinetic/pharmacodynamic relationship

At a dose of memantine of 20 mg per day the CSF levels match the k_i -value (k_i = inhibition constant) of memantine, which is 0.5 μ mol in human frontal cortex.

Pharmacodynamics:

Pharmacotherapeutic group: Psychoanaleptics. Other Anti-dementia drugs, ATC code: N06DX01.

There is increasing evidence that malfunctioning of glutamatergic neurotransmission, in particular at NMDA-receptors, contributes to both expression of symptoms and disease

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MACMENTA 20 (MEMANTINE HYDROCHLORIDE 20 MG FILM-COATED TABLETS)

Macleods Pharmaceuticals Ltd.

progression in neurodegenerative dementia. Memantine is a voltage-dependent, moderate-affinity uncompetitive NMDA-receptor antagonist. It modulates the effects of pathologically elevated tonic levels of glutamate that may lead to neuronal dysfunction.

INDICATIONS AND USAGE:

Treatment of patients with moderate to severe Alzheimer's disease.

CONTRAINDICATIONS:

Hypersensitivity to the active substance or to any of the excipients of Memantine hydrochloride.

ADVERSE REACTION:

Summary of the safety profile

The most frequently occurring adverse reactions were dizziness, headache, constipation, somnolence and hypertension.

Tabulated list of adverse reactions

Adverse reactions are ranked according to system organ class, using the following convention: very common, common, uncommon, rare, very rare, not known. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

<i>SYSTEM ORGAN CLASS</i>	<i>FREQUENCY</i>	<i>ADVERSE REACTION</i>
Infections and infestations	Uncommon	Fungal infections
Immune system disorders	Common	Drug hypersensitivity
Psychiatric disorders	Common	Somnolence
	Uncommon	Confusion
	Uncommon	Hallucinations ¹
	Not known	Psychotic reactions ²
Nervous system disorders	Common	Dizziness
	Common	Balance disorders
	Uncommon	Gait abnormal

MACMENTA 10 (MEMANTINE HYDROCHLORIDE 10 MG FILM-COATED TABLETS)
MACMENTA 20 (MEMANTINE HYDROCHLORIDE 20 MG FILM-COATED TABLETS)

	Very rare	Seizures
Cardiac disorders	Uncommon	Cardiac failure
Vascular disorders	Common	Hypertension
	Uncommon	Venous thrombosis/thromboembolism
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea
Gastrointestinal disorders	Common	Constipation
	Uncommon	Vomiting
	Not known	Pancreatitis ²
Hepatobiliary disorders	Common	Elevated liver function test
	Not known	Hepatitis
General disorders and administration site conditions	Common	Headache
	Uncommon	Fatigue

¹ Hallucinations have mainly been observed in patients with severe Alzheimer's disease.

² Isolated cases reported in post-marketing experience.

Alzheimer's disease has been associated with depression, suicidal ideation and suicide. In post-marketing experience these reactions have been reported in patients treated with Memantine Tablets

DOSAGE:

Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia. Therapy should only be started if a caregiver is available who will regularly monitor the intake of the medicinal product by the patient. Diagnosis should be made according to current guidelines.

The tolerance and dosing of memantine should be reassessed on a regular basis, preferably within three months after start of treatment. Thereafter, the clinical benefit of memantine and the patient's tolerance of treatment should be reassessed on a regular basis according to

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current clinical guidelines. Maintenance treatment can be continued for as long as a therapeutic benefit is favourable and the patient tolerates treatment with memantine. Discontinuation of memantine should be considered when evidence of a therapeutic effect is no longer present or if the patient does not tolerate treatment.

Memantine should be administered once a day and should be taken at the same time every day. The film-coated tablets can be taken with or without food.

Adults:

Dose titration

The maximum daily dose is 20 mg per day. In order to reduce the risk of undesirable effects the maintenance dose is achieved by upward titration of 5 mg per week over the first 3 weeks as follows:

Week 1 (day 1-7): The patient should take half a 10 mg film-coated tablet (5 mg) per day for 7 days.

Week 2 (day 8-14): The patient should take one 10 mg film-coated tablet (10 mg) per day for 7 days.

Week 3 (day 15-21): The patient should take one and a half 10 mg film-coated tablets (15 mg) per day for 7 days.

From Week 4 on: The patient should take 20mg film-coated tablet per day.

Maintenance dose: The recommended maintenance dose is 20 mg per day.

Elderly:

The recommended dose for patients over the age of 65 years is 20 mg per day (or two 10 mg film-coated tablets once a day) as described above.

MACMENTA 10 (MEMANTINE HYDROCHLORIDE 10 MG FILM-COATED TABLETS)
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Children and adolescents:

Memantine is not recommended for use in children below 18 years due to a lack of data on safety and efficacy.

Renal impairment: In patients with mildly impaired renal function (creatinine clearance 50-80 ml/min) no dose adjustment is required. In patients with moderate renal impairment (creatinine clearance 30-49 ml/min) daily dose should be 10 mg per day. If tolerated well after at least 7 days of treatment, the dose could be increased up to 20 mg/day according to standard titration scheme. In patients with severe renal impairment (creatinine clearance 5-29 ml/min) daily dose should be 10 mg per day.

Hepatic impairment: In patients with mild or moderate hepatic impaired function (Child-Pugh A and Child-Pugh B), no dose adjustment is needed. No data on the use of memantine in patients with severe hepatic impairment are available. Administration of Memantine is not recommended in patients with severe hepatic impairment.

ROUTE OF ADMINISTRATION: Oral

DRUG INTERACTIONS:

Due to the pharmacological effects and the mechanism of action of memantine the following interactions may occur:

- The mode of action suggests that the effects of L-dopa, dopaminergic agonists, and anticholinergics may be enhanced by concomitant treatment with NMDA-antagonists such as memantine. The effects of barbiturates and neuroleptics may be reduced. Concomitant administration of memantine with the antispasmodic agents, dantrolene or baclofen, can modify their effects and a dose adjustment may be necessary.
- Concomitant use of memantine and amantadine should be avoided, owing to the risk of pharmacotoxic psychosis. Both compounds are chemically related NMDA-antagonists. The same may be true for ketamine and dextromethorphan (see also section WARNING AND PRECAUTION). There is one published case report on a possible risk also for the combination of memantine and phenytoin.

MACMENTA 10 (MEMANTINE HYDROCHLORIDE 10 MG FILM-COATED TABLETS)
MACMENTA 20 (MEMANTINE HYDROCHLORIDE 20 MG FILM-COATED TABLETS)

- Other active substances such as cimetidine, ranitidine, procainamide, quinidine, quinine and nicotine that use the same renal cationic transport system as amantadine may also possibly interact with memantine leading to a potential risk of increased plasma levels.
- There may be a possibility of reduced serum level of hydrochlorothiazide (HCT) when memantine is co-administered with HCT or any combination with HCT.
- In post-marketing experience, isolated cases with international normalized ratio (INR) increases have been reported in patients concomitantly treated with warfarin. Although no causal relationship has been established, close monitoring of prothrombin time or INR is advisable for patients concomitantly treated with oral anticoagulants.

No relevant active substance-active substance interaction of memantine with glyburide/metformin or donepezil was observed.

No relevant effect of memantine on the pharmacokinetics of galantamine was observed.

Memantine did not inhibit CYP 1A2, 2A6, 2C9, 2D6, 2E1, 3A, flavin containing monooxygenase, epoxide hydrolase or sulphation *in vitro*.

WARNING & PRECAUTION:

Caution is recommended in patients with epilepsy, former history of convulsions or patients with predisposing factors for epilepsy.

Concomitant use of N-methyl-D-aspartate (NMDA)-antagonists such as amantadine, ketamine or dextromethorphan should be avoided. These compounds act at the same receptor system as memantine, and therefore adverse reactions (mainly central nervous system (CNS)-related) may be more frequent or more pronounced (see also section DRUG INTERACTIONS).

Some factors that may raise urine pH (see section Pharmacokinetic “Elimination”) may necessitate careful monitoring of the patient. These factors include drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or a massive ingestion of alkalisating gastric buffers. Also, urine pH may be elevated by states of renal tubular acidosis (RTA) or severe infections of the urinary tract with *Proteus bacteria*.

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Patients with recent myocardial infarction, uncompensated congestive heart failure (NYHA III-IV), or uncontrolled hypertension should be closely supervised.

PREGNANCY AND LACTATION:

Pregnancy

There are no or limited amount of data from the use of memantine in pregnant women. The potential risk for humans is unknown. Memantine should not be used during pregnancy unless clearly necessary.

Breast-feeding

It is not known whether memantine is excreted in human breast milk but, taking into consideration the lipophilicity of the substance, this probably occurs. Women taking memantine should not breast-feed.

Fertility

No adverse reactions of memantine were noted on male and female fertility.

OVERDOSE:

Only limited experience with overdose is available.

Symptoms

Relative large overdoses (200 mg and 105 mg/day for 3 days, respectively) have been associated with either only symptoms of tiredness, weakness and/or diarrhoea or no symptoms. In the overdose cases below 140 mg or unknown dose the patients revealed symptoms from central nervous system (confusion, drowsiness, somnolence, vertigo, agitation, aggression, hallucination, and gait disturbance) and/or of gastrointestinal origin (vomiting and diarrhoea).

In the most extreme case of overdose, the patient survived the oral intake of a total of 2000 mg memantine with effects on the central nervous system (coma for 10 days, and later diplopia and agitation). The patient received symptomatic treatment and plasmapheresis. The patient recovered without permanent sequelae.

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In another case of a large overdose, the patient also survived and recovered. The patient had received 400 mg memantine orally. The patient experienced central nervous system symptoms such as restlessness, psychosis, visual hallucinations, proconvulsiveness, somnolence, stupor, and unconsciousness.

Treatment

In the event of overdose, treatment should be symptomatic. No specific antidote for intoxication or overdose is available. Standard clinical procedures to remove active substance material, e.g. gastric lavage, carbo medicinalis (interruption of potential entero-hepatic recirculation), acidification of urine, forced diuresis should be used as appropriate. In case of signs and symptoms of general central nervous system (CNS) overstimulation, careful symptomatic clinical treatment should be considered.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Moderate to severe Alzheimer's disease usually causes impairment of driving performance and compromises the ability to use machinery. Furthermore, Memantine Tablets has minor to moderate influence on the ability to drive and use machines such that outpatients should be warned to take special care.

STORAGE:

Store below 30°C in a dry place. Protect from light.

PRESENTATION:

Alu / Clear PVC/PVdC Blister of 14 Tablets, Such 4 blisters packed in a carton along with the Pack Insert.

MANUFACTURED BY

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Macleods Pharmaceuticals Ltd.

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Product Registration Holder in Malaysia

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